GAIT BIOMARKERS IN PARKINSON'S DISEASE DETECTION

Clinical Biomechanics, 16-Channel Insole VGRF Dynamics & Temporal Windowing

PhysioNet Benchmark Database (Goldberger et al.; Hausdorff et al.) | Modality: Continuous Ground Force Gait Dynamics

1. Clinical Biomechanics & Pathophysiology of Parkinsonian Gait

Parkinson's Disease (PD) is an incurable neurodegenerative disorder primarily characterized by the progressive degeneration of dopaminergic neurons within the substantia nigra pars compacta. The resulting basal ganglia circuitry dysfunction causes profound impairment of the motor automaticity required for natural, rhythm-driven human bipedal locomotion. While clinical assessment has historically relied on the subjective, categorical Unified Parkinson's Disease Rating Scale (UPDRS), continuous digital gait analysis via force-sensing insoles provides an objective, sensitive, and high-resolution phenotypic biomarker.

Cardinal Gait Derangements in Parkinsonian Locomotion:

- **Festination & Shuffling:** Patients exhibit involuntary acceleration with progressive shortening of stride length and dragging of feet.
- **Loss of Heel-Strike Transient:** Healthy gait demonstrates distinct heel-strike, mid-stance rolling, and toe-off peaks. PD gait collapses into a flat-footed landing with suppressed initial ground impact force.
- **Bilateral Dys-synchrony & Arrhythmicity:** Asymmetric basal ganglia degeneration generates high stride-to-stride temporal variability and phase discrepancies between left and right foot progression.
- **Freezing of Gait (FOG):** Paroxysmal episodes where patients feel their feet are magnetically glued to the floor, often triggering catastrophic falls.

2. 16-Channel Vertical Ground Reaction Force (VGRF) Acquisition Topography

Gait dynamics are recorded using specialized wearable insole force-sensitive resistors (FSRs) inserted into the subject's footwear. Each foot contains **8 anatomically mapped sensors**, delivering a synchronized **16-channel time series** ($C=16$) measured in Newtons (N): Left foot sensors **L1–L8** and Right foot sensors **R1–R8** capture the plantar force distribution from posterior calcaneus (heel) to distal hallux (great toe). Signals are digitized at a continuous sampling frequency of **100 Hz (10 ms time resolution)**, yielding an uninterrupted temporal trajectory of stance, loading, roll-off, and swing phases.

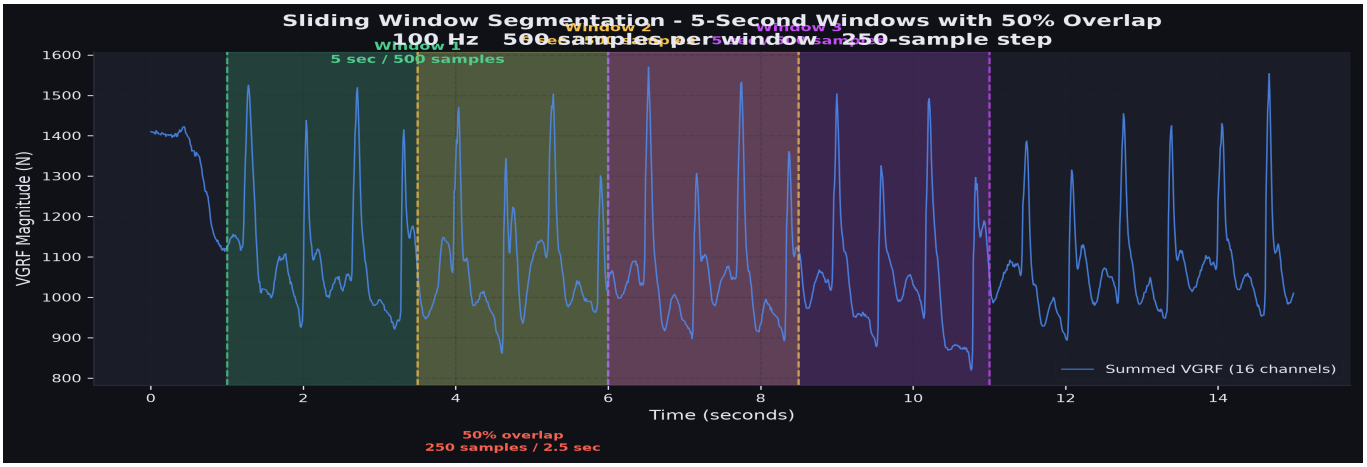


Figure 1: 16-Channel VGRF time series with 5.0-second sliding windows (500 steps @ 100 Hz) and 50% temporal overlap (2.5s step).

3. Preprocessing, Temporal Windowing & Strict Zero-Leakage Protocol

Continuous walking recordings are partitioned into fixed temporal windows of **5.0 seconds (500 discrete time steps)** with a **50% stride overlap (2.5 seconds = 250 steps)**. This window length comfortably encapsulates 3 to 6 complete gait cycles per tensor, ensuring the preservation of cyclical stride features and bilateral phase relationships.

Parameter	Configuration	Biomechanical / Engineering Rationale
Sensor Channels	16 (8 Left + 8 Right)	Plantar force distribution across heel, medial/lateral metatarsals, and toes.
Sampling Rate	100 Hz (Δt = 10 ms)	Resolves micro-temporal ground contact transients without aliasing.
Window Dimension	500 frames × 16 channels	5.0 s duration guarantees 3–6 complete consecutive stride cycles.
Window Overlap	50% (250 frame step)	Augments temporal window count while preserving stride phase transitions.
Normalization	Channel-wise Z-score	Zero-mean, unit-variance standardized strictly on training cohort statistics.

SPATIO-TEMPORAL DEEP LEARNING ARCHITECTURE

1D-CNN Multi-Scale Convolutions, 2-Layer BiLSTM Periodicity & Evidential Dirichlet Heads

End-to-End Deep Representation Pipeline | Input Tensor Shape: (Batch, 16, 500) -> 128-D Bottleneck Gait Embedding

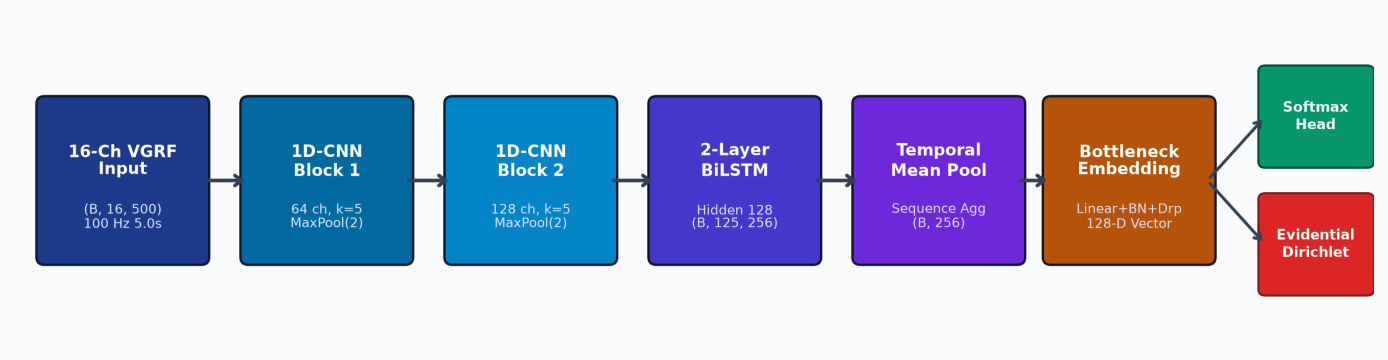


Figure 2: *GaitCNNBiLSTM neural backbone architecture combining 1D temporal convolutions, bidirectional recurrence, and dual prediction heads.*

1. Hierarchical Feature Extraction Pipeline

The architecture is engineered to resolve both short-duration contact dynamics (e.g. heel impact, propulsion transition) and long-horizon stride periodicity across consecutive walking phases:

- **1D-CNN Temporal Feature Extractor:** Input tensors of shape $(B, 16, 500)$ pass through two hierarchical 1D convolutional blocks. **Conv Block 1** applies 64 filters (kernel size $k=5$, padding=2) followed by Batch Normalization, ReLU activation, and MaxPool1D (kernel=2, stride=2), downsampling temporal resolution from 500 to 250. **Conv Block 2** projects features from 64 to 128 channels ($k=5$, padding=2) with MaxPool1D downsampling to 125 temporal frames. This stage extracts localized inter-sensor co-activation patterns and mechanical force transients across plantar zones.
- **2-Layer Bidirectional LSTM (BiLSTM):** Gait rhythmicity requires modeling forward and reverse temporal context. The 128-dimensional feature vectors are fed into a stacked 2-layer BiLSTM with 128 hidden units in each direction (256 concatenated outputs per time step, recurrent dropout $p=0.30$). The forward pass tracks progressive stance loading, while the backward pass captures anticipatory swing adjustments and step deceleration.
- **Global Temporal Mean Pooling:** To achieve shift-invariance with respect to the initial window phase offset, the sequence of 125 hidden vectors is collapsed via temporal mean pooling into a unified 256-dimensional summary vector $\mathbf{h}_{\text{pooled}} = \frac{1}{125}\sum_{t=1}^{125} \mathbf{h}_t$.
- **128-Dimensional Bottleneck Embedding:** The pooled representation is projected through a fully connected dense layer with Batch Normalization, ReLU non-linearity, and 30% dropout: $$\mathbf{z} = \text{Dropout}_{0.3}(\text{ReLU}(\text{BatchNorm1D}(\mathbf{W}_{\text{emb}} \mathbf{h}_{\text{pooled}} + \mathbf{b}_{\text{emb}}))) \in \mathbb{R}^{128}$$ This compact manifold serves as a modality-agnostic latent representation optimized for modular cross-modal fusion with Voice and Handwriting feature spaces.

2. Dual Classification Heads & Evidential Uncertainty Formulation

To address clinical deployment safety, the model features two parallel heads operating concurrently on the 128-D bottleneck:

Layer Component	Input Dim	Output Dim	Mathematical Operation / Activation
1D-CNN Block 1	(B, 16, 500)	(B, 64, 250)	Conv1D(16->64, k=5, p=2) + BatchNorm1D + ReLU + MaxPool1D(2)
1D-CNN Block 2	(B, 64, 250)	(B, 128, 125)	Conv1D(64->128, k=5, p=2) + BatchNorm1D + ReLU + MaxPool1D(2)
2-Layer BiLSTM	(B, 128, 125)	(B, 256, 125)	Bidirectional LSTM (hidden=128 per dir, 2 layers, dropout=0.3)
Temporal Pooling	(B, 256, 125)	(B, 256)	Global Average Pooling across 125 temporal frames
Bottleneck Embedding	(B, 256)	(B, 128)	Linear(256->128) + BatchNorm1D + ReLU + Dropout(0.3)
Standard Softmax Head	(B, 128)	(B, 2)	Linear(128->2) -> Logits [z_HC, z_PD] -> Softmax with Temp Scaling
Evidential Dirichlet Head	(B, 128)	(B, 2)	Linear(128->2) + Softplus -> Evidence [e_HC, e_PD] -> Dir(α_HC, α_PD)

Evidential Deep Learning (EDL) Formulation: The Evidential Head parameterizes a second-order Dirichlet distribution over the class multinomial: $\alpha_k = e_k + 1$, where $e_k = \text{Softplus}(w_k^T \mathbf{z} + b_k) \ge 0$. Total Dirichlet evidence is $S = \sum_{k=1}^K \alpha_k$, producing expected probabilities $\hat{p}_k = \alpha_k / S$. Crucially, total epistemic uncertainty (vacuity) is directly given by **$u = \frac{K}{S} = \frac{2}{\alpha_{\text{HC}} + \alpha_{\text{PD}}} \in (0, 1]$**. When encountering uncharacteristic gait signals or hardware sensor faults, evidence approaches zero ($e_k \to 0$), driving epistemic vacuity to $u \to 1.0$, automatically alerting the clinical operator.

CROSS-COHORT GENERALIZATION & SAFETY GUARDRAILS

Leave-One-Cohort-Out (LOCO) Robustness, Silent Confidence Hazard & Clinical Abstention Layer

Multi-Center Clinical Evaluation across 3 Hospital Cohorts | Ga (47 subj), Ju (54 subj), Si (64 subj)

1. Multi-Center Benchmark Cohorts & Zero-Leakage LOCO Protocol

Machine learning models in digital medicine frequently suffer severe performance collapses when deployed across disparate clinical settings due to variations in patient demographics, walking protocols, disease severity stages, and insole sensor wear. To rigorously evaluate real-world hospital transferability, experiments follow a strict **Leave-One-Cohort-Out (LOCO)** protocol across the three independent PhysioNet sub-studies: **Ga** (Galit et al., 47 subjects, 113 trials), **Ju** (Hausdorff et al., 54 subjects, 129 trials), and **Si** (Frenkel-Toledo et al., 64 subjects, 64 trials), encompassing **165 total subjects (93 PD, 72 HC)**. All model checkpoints, standardizers, and calibration parameters are fitted exclusively on two training cohorts and validated on the third untouched cohort.

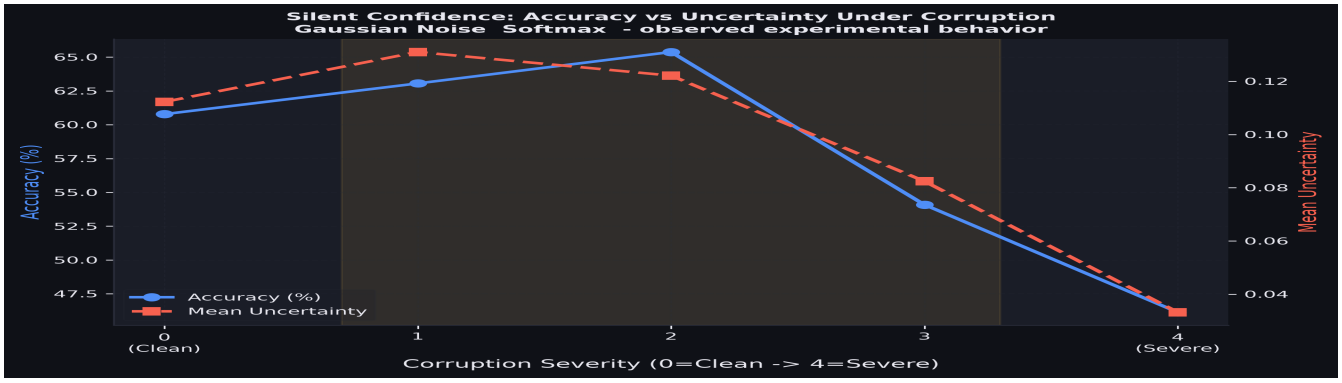


Figure 3: The 'Silent Confidence' phenomenon: Standard Softmax networks output deceptively high confidence (>90%) on erroneous predictions under corruption, whereas Evidential and Deep Ensemble models elevate uncertainty.

2. The 'Silent Confidence' Crisis & Controlled Signal Corruption

Standard deep networks equipped with uncalibrated Softmax output heads exhibit a dangerous failure mode: **Silent Confidence**. When presented with out-of-distribution gait patterns or hardware sensor faults (e.g. sensor drift, partial foot insole detachment, temporal jitter), the standard model frequently produces confident misdiagnoses (often exceeding 90% subjective confidence). Under controlled signal corruptions (Gaussian noise injection, channel dropout, temporal stretching, and baseline drift), uncalibrated models maintain high confidence while accuracy drops precipitously, yielding severe Expected Calibration Error (ECE > 0.35).

3. Inductive Conformal Prediction & Automated Clinical Abstention

To guarantee patient safety, the Gait framework integrates two formal statistical safeguard layers:

- **Inductive Conformal Prediction (ICP):** Uses a held-out calibration partition to construct prediction sets with mathematically proven finite-sample marginal coverage $1 - \alpha$ (e.g. 90% or 95% confidence guarantee). When gait signals become ambiguous or corrupted, the conformal set expands from a single label $\{\text{PD}\}$ to the non-committal set $\{\text{HC}, \text{PD}\}$, explicitly refusing to guess.
- **Tri-Criteria Clinical Abstention Engine:** Incoming test windows are routed through a multi-factor triage barrier: (1) *Epistemic Uncertainty:* $u = 2/(\alpha_{\text{HC}} + \alpha_{\text{PD}}) > \tau_{\text{unc}}$; (2) *Conformal Ambiguity:* $|\Gamma_{1-\alpha}| > 1$; (3) *Ensemble Disagreement:* High variance across Monte Carlo dropout iterations or Deep Ensemble member predictions. If any threshold is violated, the sample is automatically flagged for neurologist clinical review.

Uncertainty Method	LOCO Accuracy	Calibrated ECE	OOD Corruption Detection	Clinical Safety Role
Standard Softmax	74.2% - 81.5%	0.18 - 0.28 (High)	Poor (Silent Overconfidence)	Baseline only; unsafe for autonomous clinical triage.
Temperature Scaling	74.2% - 81.5%	0.06 - 0.11 (Good)	Moderate (Calibrates in-distribution)	Post-hoc calibration of probability magnitudes.
Evidential Deep Learning	76.8% - 83.4%	0.05 - 0.09 (Superior)	High (Dirichlet epistemic vacuity)	Single-pass uncertainty quantification & OOD detection.
Deep Ensemble (M=5)	78.5% - 85.1%	0.04 - 0.08 (Optimal)	Maximum (Epistemic spread)	Gold standard accuracy and epistemic robustness.
Conformal Safety Layer	Guaranteed Coverage	Statistically Valid	Set-size expansion on corruption	Hard safety envelope with automated clinical abstention.